CONTINUOUS K-THEORY

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Abstract. This document contains notes prepared by Yiming Wang based on the seminar "Efimov K-theory", organized by Marc Hoyois at the University of Regensburg during the winter semester 2027-2028.

The author used ChatGPT 6 for grammatical and mathematical revision. Nevertheless, any errors are solely the author's responsibility. Comments, corrections, and suggestions are welcome.

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Date: October 16, 2026.

1. Introduction

Algebraic K-theory were defined for small stable categories. However, many categories arising in nature are instead large: examples include categories of sheaves on topological spaces and categories of nuclear modules over analytic rings. Efimov's continuous K-theory extends algebraic K-theory to dualizable presentable stable categories.

Notation 1.1. Throughout these notes, we adopt the following notations:

- *categories* refer to $(\infty, 1)$-categories;
- $\mathrm{Cat}^{\mathrm{st}}$ denotes the category of small stable categories and exact functors;
- $\mathrm{Cat}^{\mathrm{st,idem}} \subseteq \mathrm{Cat}^{\mathrm{st}}$ denotes the full subcategory of small idempotent-complete stable categories;
- $\mathrm{Pr}^L_{\mathrm{st}}$ for the category of presentable stable categories and colimit-preserving functors;
- An denotes the category of animae, whereas Sp denotes the category of spectra;
- *algebraic K-theory* refer to the nonconnective K-theory functor $\mathrm{K}\colon \mathrm{Cat}^{\mathrm{st,idem}} \to \mathrm{Sp}$.

Definition 1.2 (Localizing sequences)**.** A *localizing sequence* in $\mathrm{Cat}^{\mathrm{st,idem}}$ is a *fiber sequence* $\mathcal{A} \to \mathcal{B} \to \mathcal{C}$ in $\mathrm{Cat}^{\mathrm{st,idem}}$ such that the induced functor $(\mathcal{B}/\mathcal{A})^{\mathrm{idem}} \to \mathcal{C}$ is an equivalence. Equivalently,

$$\mathrm{Ind}(\mathcal{A}) \hookrightarrow \mathrm{Ind}(\mathcal{B}) \longrightarrow \mathrm{Ind}(\mathcal{C})$$

is a *localizing sequence* in $\mathrm{Pr}^L_{\mathrm{st}}$ (see [Nee92][Theorem 2.1]), i.e., it is both a fiber sequence and a cofiber sequence in $\mathrm{Pr}^L_{\mathrm{st}}$. The adjoints of these functors give the associated stable recollement.

Definition 1.3 (Localizing invariants)**.** Let $F\colon \mathrm{Cat}^{\mathrm{st,idem}} \to \mathrm{Sp}$ be a functor with $F(0) \simeq 0$. The following conditions are equivalent:

(a) For every fully faithful exact functor $\mathcal{A} \hookrightarrow \mathcal{B}$, the sequence

$$F(\mathcal{A}) \longrightarrow F(\mathcal{B}) \longrightarrow F(\mathcal{B}/\mathcal{A})$$

is a fiber sequence.

(b) F carries every localizing sequence to a fiber sequence.

In this case, the functor F is called a *localizing invariant*. We denote the *category of localizing invariants* by $\mathrm{Fun}_{\mathrm{loc}}(\mathrm{Cat}^{\mathrm{st,idem}}, \mathrm{Sp})$.

The localization theorem for algebraic K-theory says that K is a localizing invariant. It also has the following universal property.

Theorem 1.4 (Universal property of K-theory)**.** *The functor* K *is the unit for the localized Day convolution on* $\mathrm{Fun}_{\mathrm{loc}}(\mathrm{Cat}^{\mathrm{st,idem}}, \mathrm{Sp})$. *The canonical transformation* $\Sigma^\infty_+ \mathrm{Core} \to \mathrm{K}$ *induces an equivalence*

$$\mathrm{Map}(\mathrm{K}, F) \xrightarrow{\sim} \mathrm{Map}(\Sigma^\infty_+ \mathrm{Core}, F)$$

for every localizing invariant F. Equivalently, every natural transformation $\Sigma^\infty_+ \mathrm{Core} \to F$ extends to a map $\mathrm{K} \to F$, unique up to a contractible space of choices, making the diagram commute:

$$\begin{array}{ccc} \Sigma^\infty_+ \mathrm{Core} & \longrightarrow & F. \\ \downarrow & \nearrow_{\exists!} & \\ K & & \end{array}$$

To relate the two formulations, consider the inclusions

$$\mathrm{Fun}_{\mathrm{loc}}(\mathrm{Cat}^{\mathrm{st,idem}}, \mathrm{Sp}) \subseteq \mathrm{Fun}(\mathrm{Cat}^{\mathrm{st,idem}}, \mathrm{Sp}) \subseteq \mathrm{Fun}(\mathrm{Cat}^{\mathrm{st,idem}}, \widehat{\mathrm{Sp}}),$$

where $\widehat{\mathrm{Sp}}$ allows a larger universe. Consider the suspension-spectrum Yoneda construction

$$\mathrm{Cat}^{\mathrm{st,idem}} \to \mathrm{Fun}(\mathrm{Cat}^{\mathrm{st,idem}}, \mathrm{Sp}), \quad \mathcal{A} \mapsto \Sigma^\infty_+ \mathrm{Map}(\mathcal{A}, -)$$

Since the tensor unit of $\mathrm{Cat}^{\mathrm{st,idem}}$ is the category $\mathrm{Sp}^{\mathrm{fin}}$ of finite spectra, the unit for Day convolution is

$$\Sigma^\infty_+ \operatorname{Map}(\mathrm{Sp}^{\mathrm{fin}}, -) \simeq \Sigma^\infty_+ \operatorname{Core}.$$

Its image under localization to localizing invariants is K.

The following equivalence it the passage from small categories to presentable categories:

$$\operatorname{Ind}\colon \mathrm{Cat}^{\mathrm{st,idem}} \xrightarrow{\sim} \mathrm{Pr}^L_{\omega,\mathrm{st}},$$

where $\mathrm{Pr}^L_{\omega,\mathrm{st}}$ consists of compactly generated presentable stable categories and colimit-preserving functors that also preserve compact objects; the inverse sends $\mathcal{C}$ to $\mathcal{C}^\omega$.

There is an asymmetry between fully faithful functors and localizing sequences under this equivalence:

Remark 1.5 (An asymmetry). Let $f\colon \mathcal{A} \to \mathcal{B}$ be an exact functor in $\mathrm{Cat}^{\mathrm{st,idem}}$.

(a) f is fully faithful if and only if $\operatorname{Ind}(f)$ is fully faithful. In $\mathrm{Pr}^L_{\mathrm{st}}$, the latter condition is equivalent to being a monomorphism.

(b) If $\ker(f) \hookrightarrow \mathcal{A} \xrightarrow{f} \mathcal{B}$ is a localizing sequence, then $\operatorname{Ind}(f)$ is a localization, or equivalently an epimorphism in $\mathrm{Pr}^L_{\mathrm{st}}$. The converse need not hold.

An exact functor whose Ind-completion is a localization is often called a *homological epimorphism*. Its kernel after Ind-completion need not be generated by compact objects.

Theorem 1.6 (Tamme)**.** *Consider a commutative square Q of small idempotent-complete stable categories*

$$\begin{array}{ccc} \mathcal{A} & \xrightarrow{f} & \mathcal{B} \\ \downarrow & & \downarrow \\ \mathcal{C} & \xrightarrow{g} & \mathcal{D}. \end{array}$$

Suppose that $\operatorname{Ind}(Q)$ *is cartesian and that* $\operatorname{Ind}(g)$ *is a localization. Then $F(Q)$ is cartesian for every localizing invariant F. Under these hypotheses, Q is itself cartesian and* $\operatorname{Ind}(f)$ *is also a localization.*

Remark 1.7 (A tempting but incorrect proof). Since Q is cartesian, there is an equivalence $\ker(f) \simeq \ker(g)$. As F takes value in a stable category, one might therefore try to apply F to the two sequences

$$\ker(f) \longrightarrow \mathcal{A} \longrightarrow \mathcal{B}, \qquad \ker(g) \longrightarrow \mathcal{C} \longrightarrow \mathcal{D},$$

and compare their fibers. By Remark 1.5, the gap is that these need not be localizing sequences of small categories.

Example 1.8 (Milnor patching). Consider a cartesian square of $\mathbf{E}_1$-rings

$$\begin{array}{ccc} A & \longrightarrow & A' \\ \downarrow & & \downarrow \\ B & \longrightarrow & B' \end{array}$$

and suppose that $A \to A'$ is *Tor-unital*[1], i.e., the multiplication map $A' \otimes_A A' \xrightarrow{\sim} A'$ is an isomorphism. In this case, $B \to B'$ is also Tor-unital. Extension of scalars then gives a cartesian square

$$\begin{array}{ccc} \mathrm{Mod}_A & \longrightarrow & \mathrm{Mod}_{A'} \\ \downarrow & & \downarrow \\ \mathrm{Mod}_B & \longrightarrow & \mathrm{Mod}_{B'}, \end{array}$$

[1]This condition guarantees that the extension of scalar $\mathrm{Mod}_A \to \mathrm{Mod}_{A'}$ is a localization.

where both horizontal functors are localizations. Since $\mathrm{Mod}_A \simeq \mathrm{Ind}(\mathrm{Perf}_A)$. Since $\mathrm{K}(A) = \mathrm{K}(\mathrm{Perf}_A)$ is a localizing invariant, Theorem 1.6 applies and we obtain a cartesian square

$$\begin{array}{ccc} \mathrm{K}(A) & \longrightarrow & \mathrm{K}(A') \\ \downarrow & & \downarrow \\ \mathrm{K}(B) & \longrightarrow & \mathrm{K}(B'). \end{array}$$

This is one of the most significant results in modern algebraic K-theory.

However, a cartesian square of ordinary rings does not automatically satisfy the Tor-unitality hypothesis:

Example 1.9 (Classical Milnor squares)*.* Consider a cartesian square of ordinary rings

$$\begin{array}{ccc} A & \longrightarrow & A/I \\ \downarrow & & \downarrow \\ B & \longrightarrow & B/IB. \end{array}$$

Although the ordinary tensor product $(A/I) \otimes_A (A/I)$ is A/I, the derived tensor product can have nonzero higher homotopy groups. For example, the first homotopy group is

$$\mathrm{Tor}_1^A(A/I, A/I) \simeq I/I^2.$$

Thus algebraic K-theory need not carry the cartesian square to a cartesian square.

Keller's example makes the obstruction in Remark 1.5 particularly explicit.

Example 1.10 (Keller's example)*.* Let A be a commutative ring and let $I \subseteq \mathrm{Jac}(A)$ be an ideal such that $A \to A/I$ is Tor-unital. The functor $\mathrm{Mod}_A \longrightarrow \mathrm{Mod}_{A/I}$ is a localization, but its kernel contains no nonzero object that is compact in Mod_A. Indeed, a perfect complex P with $P \otimes_A A/I \simeq 0$ vanishes by derived Nakayama. [2] Consequently, the kernel has no nonzero compact objects of its own.

For a concrete example, let k be a field and

$$A = k[t, t^{1/2}, t^{1/4}, \ldots]_{(t, t^{1/2}, t^{1/4}, \ldots)}, \qquad \mathfrak{m} = (t, t^{1/2}, t^{1/4}, \ldots).$$

This is a local ring, whose maximal ideal $\mathfrak{m}$ is flat and satisfies $\mathfrak{m}^2 = \mathfrak{m}$, so $A \to A/\mathfrak{m}$ is Tor-unital. The functor

$$0 \hookrightarrow \mathrm{Perf}_A \longrightarrow \mathrm{Perf}_{A/\mathfrak{m}}$$

is not a localizing sequence, although the Ind-completion $\mathrm{Mod}_A \longrightarrow \mathrm{Mod}_{A/I}$ is a localization. Indeed, the functor above is not fully faithful. Such examples are a basic feature of *almost mathematics*.

Theorem 1.11 (Characterizations of dualizability)**.** *Let $\mathcal{C} \in \mathrm{Pr}^L_{\mathrm{st}}$. The following conditions are equivalent:*

(a) there is a localization $\varphi\colon \mathcal{D} \to \mathcal{E}$ in $\mathrm{Pr}^L_{\omega,\mathrm{st}}$ such that $\mathcal{C} \simeq \ker(\varphi)$;
(b) $\mathcal{C}$ is dualizable in the symmetric monoidal category $(\mathrm{Pr}^L_{\mathrm{st}}, \otimes)$;
(c) the colimit functor $\mathrm{Ind}(\mathcal{C}) \to \mathcal{C}$ has a left adjoint, i.e., $\mathcal{C}$ is compactly assembled;
(d) $\mathcal{C}$ is a retract in $\mathrm{Pr}^L_{\mathrm{st}}$ of a compactly generated category;
(e) (Efimov) *the functor $\mathcal{C} \otimes -$ preserves fully faithful colimit-preserving functors between presentable stable categories, i.e., $\mathcal{C}$ is* flat*;*
(f) (Clausen–Scholze) *$\mathcal{C}$ satisfies* AB6*: for every family of filtered categories $(J_i)_{i\in I}$ and diagrams $X_i\colon J_i \to \mathcal{C}$, the canonical map*

$$\operatorname*{colim}_{(j_i)\in\prod_{i\in I} J_i} \prod_{i\in I} X_i(j_i) \longrightarrow \prod_{i\in I} \operatorname*{colim}_{j_i\in J_i} X_i(j_i)$$

is an equivalence.

[2] The right adjoint of this localization preserves colimits, so the inclusion of its kernel also has a colimit-preserving right adjoint, hence the inclusion preserves compact objects.

Example 1.12 (Dualizable categories).

(a) The kernel in Example 1.10 is dualizable, although it is not compactly generated.
(b) If X is a locally compact Hausdorff space, then the *category of sheaves* $\mathrm{Shv}(X, \mathrm{Sp})$ is dualizable.
(c) Categories of *nuclear modules* provide further examples:

$$\mathrm{Nuc}(\mathbf{Z}_p^{\blacksquare}), \qquad \mathrm{Nuc}(\mathbf{R}^{p\text{-liq}}), \qquad \mathrm{Nuc}(\mathbf{C}^{\mathrm{gas}}),$$

where $0 < p \leq 1$.

The categories in (b) and (c) admit natural self-dualities.

For symmetric monoidal categories, rigidity gives a useful source of self-duality:

Definition 1.13 (Rigid categories)**.** A category $\mathcal{C} \in \mathrm{CAlg}(\mathrm{Pr}^L_{\mathrm{st}})$ is *rigid* if the multiplication functor

$$\mu_I \colon \mathcal{C}^{\otimes I} \longrightarrow \mathcal{C}$$

has a right adjoint in $\mathrm{Mod}_{\mathcal{C}^{\otimes I}}(\mathrm{Pr}^L_{\mathrm{st}})$ for every finite set I.

Proposition 1.14. *A rigid category $\mathcal{C}$ is self-dual in $\mathrm{Pr}^L_{\mathrm{st}}$. Its evaluation pairing is*

$$\mathcal{C} \otimes \mathcal{C} \xrightarrow{\mu} \mathcal{C} \xrightarrow{\Gamma} \mathrm{Sp}, \qquad \Gamma = \mathrm{map}_{\mathcal{C}}(\mathbf{1}_{\mathcal{C}}, -),$$

where $\mathrm{map}_{\mathcal{C}}$ is the mapping spectrum.

Definition 1.15 (The category of dualizable categories)**.** We denote by $\mathrm{Cat}^{\mathrm{dual}}_{\mathrm{st}}$ the subcategory of $\mathrm{Pr}^L_{\mathrm{st}}$ whose objects are dualizable categories and whose morphisms are colimit-preserving functors with colimit-preserving right adjoints.[3] Moreover, $\mathrm{Cat}^{\mathrm{dual}}_{\mathrm{st}} \subseteq \mathrm{Pr}^L_{\mathrm{st}}$ is not a full subcategory.

Remark 1.16. The Ind-completion gives a fully faithful functor

$$\mathrm{Ind} \colon \mathrm{Cat}^{\mathrm{st,idem}} \hookrightarrow \mathrm{Cat}^{\mathrm{dual}}_{\mathrm{st}}$$

whose essential image consists of the compactly generated categories.

Let $\mathcal{C} \in \mathrm{Pr}^L_{\mathrm{st}}$ be a dualizable category. By Theorem 1.11(a), we can choose a localization sequence

$$\mathcal{C} \hookrightarrow \mathcal{D} \longrightarrow \mathcal{E},$$

where $\mathcal{D}$ and $\mathcal{E}$ are compactly generated and both functors strongly continuous. Any extension $\mathrm{K}^{\mathrm{cont}}$ of K that is a *localizing invariant* must therefore have

$$\mathrm{K}^{\mathrm{cont}}(\mathcal{C}) \simeq \mathrm{fib}\big(\mathrm{K}(\mathcal{D}^{\omega}) \longrightarrow \mathrm{K}(\mathcal{E}^{\omega})\big).$$

To obtain a definition independence of such choice and functorial in $\mathcal{C}$, we use a canonical localization sequence.

Let κ be an uncountable regular cardinal. By Theorem 1.11, every dualizable category is ω_1-presentable, and the colimit functor

$$\mathrm{colim} \colon \mathrm{Ind}(\mathcal{C}^{\kappa}) \longrightarrow \mathcal{C}$$

has a fully faithful left adjoint j_{κ}. The *continuous κ-Calkin category* is defined as

$$\mathrm{Calk}^{\mathrm{cont}}_{\kappa}(\mathcal{C}) = \big(\mathrm{Ind}(\mathcal{C}^{\kappa})/j_{\kappa}(\mathcal{C})\big)^{\omega},$$

where the quotient $\mathcal{C}^{\kappa}/j_{\kappa}(\mathcal{C})$ is compactly generated. Therefore, there are adjunctions

$$\mathcal{C} \underset{\mathrm{colim}}{\overset{j_{\kappa}}{\rightleftarrows}} \mathrm{Ind}(\mathcal{C}^{\kappa}) \underset{q^R}{\overset{q}{\rightleftarrows}} \mathrm{Ind}(\mathrm{Calk}^{\mathrm{cont}}_{\kappa}),$$

where $j_{\kappa} \dashv \mathrm{colim}$ and $q \dashv q^R$. This gives us a localization sequence in $\mathrm{Cat}^{\mathrm{dual}}_{\mathrm{st}}$.

In particular, it always suffices to take $\kappa = \omega_1$.

[3] Such functors are called *strongly continuous*, or *compact*, functors. Equivalently, they are left adjoint morphisms in the 2-category $\mathrm{Pr}^L_{\mathrm{st}}$.

Remark 1.17 (Continuous localizing invariants). Let $F\colon \mathrm{Cat}^{\mathrm{st,idem}} \to \mathrm{Sp}$ be a localizing invariant. Its *continuous extension* is defined by

$$F^{\mathrm{cont}}(\mathcal{C}) = \mathrm{fib}\big(F(\mathcal{C}^{\omega_1}) \to F(\mathrm{Calk}^{\mathrm{cont}}_{\omega_1}(\mathcal{C}))\big).$$

Since $\mathcal{C}^{\omega_1}$ admits countable coproducts, the *Eilenberg swindle* gives $F(\mathcal{C}^{\omega_1}) \simeq 0$. In particular,

$$F^{\mathrm{cont}}(\mathcal{C}) \simeq \Omega F(\mathrm{Calk}^{\mathrm{cont}}_{\omega_1}(\mathcal{C})).$$

Furthermore, this construction is independent of the regular cardinal κ. For $F = \mathrm{K}$, we obtain *continuous K-theory*, denoted by $\mathrm{K}^{\mathrm{cont}}$.

Theorem 1.18 (Efimov)**.** *Restriction along* Ind *induces an equivalence of categories*

$$\mathrm{Ind}^*\colon \mathrm{Fun}_{\mathrm{loc}}(\mathrm{Cat}^{\mathrm{dual}}_{\mathrm{st}}, \mathrm{Sp}) \xrightarrow{\sim} \mathrm{Fun}_{\mathrm{loc}}(\mathrm{Cat}^{\mathrm{st,idem}}, \mathrm{Sp}),$$

where the inverse is given by $F \mapsto F^{\mathrm{cont}}$*. In particular,*

$$F^{\mathrm{cont}}(\mathrm{Ind}(\mathcal{A})) \simeq F(\mathcal{A}) \quad \textit{and} \quad \mathrm{K}^{\mathrm{cont}}(\mathrm{Ind}(\mathcal{A})) \simeq \mathrm{K}(\mathcal{A}),$$

where $\mathcal{A} \in \mathrm{Cat}^{\mathrm{st,idem}}$*.*

We conclude with examples and computations of continuous invariants.

Example 1.19 (Topological hochschild homology). *Topological hochschild homology* extends to dualizable categories by taking the trace of the identity:

$$\mathrm{THH}^{\mathrm{cont}}\colon \mathrm{Cat}^{\mathrm{dual}}_{\mathrm{st}} \longrightarrow \mathrm{Sp}, \quad \mathcal{C} \longmapsto \mathrm{tr}(\mathrm{id}_{\mathcal{C}}),$$

where the trace is an object of $\mathrm{End}_{\mathrm{Pr}^L_{\mathrm{st}}}(\mathrm{Sp}) \simeq \mathrm{Sp}$. This is a localizing invariant by the additivity theorem for traces.

Remark 1.20. Let $\mathcal{C}$ be a symmetric monoidal 2-category whose mapping categories are stable and whose composition and tensor product are exact in each variable. Generally, traces carry localization sequences of dualizable objects and right-adjointable morphisms to cofiber sequences.

Theorem 1.21 (Efimov)**.** *Let* $\mathcal{C} \in \mathrm{Cat}^{\mathrm{dual}}_{\mathrm{st}}$ *and let* X *be a locally compact Hausdorff space. There is a natural equivalence*

$$\mathrm{K}^{\mathrm{cont}}(\mathrm{Shv}(X; \mathcal{C})) \simeq \Gamma_{\mathrm{c}}\big(X; \mathrm{K}^{\mathrm{cont}}(\mathcal{C})\big),$$

where the right-hand side is the compactly supported sections of the constant sheaf of spectra.

Example 1.22 (Sheaves on Euclidean space). For every $n \geq 0$, Theorem 1.21 gives

$$\mathrm{K}^{\mathrm{cont}}(\mathrm{Shv}(\mathbf{R}^n, \mathcal{C})) \simeq \Omega^n \mathrm{K}^{\mathrm{cont}}(\mathcal{C}),$$

where $\mathcal{C} \in \mathrm{Cat}^{\mathrm{dual}}_{\mathrm{st}}$.

Theorem 1.23 (Efimov)**.** *Let* R *be a commutative ring and* $I = (f_1, \ldots, f_m) \subseteq R$ *a finitely generated ideal. Then*

$$\mathrm{K}^{\mathrm{cont}}\big(\mathrm{Nuc}(R^{\wedge}_I)\big) \simeq \lim_{n \geq 1} \mathrm{K}\big(R//(f_1^n, \ldots, f_m^n)\big),$$

where $R//(f_1^n, \ldots, f_m^n)$ *is the derived quotient.*[4]

[4]The formula holds for both Efimov's category of nuclear modules and the original Clausen–Scholze category.

2. Compactly assembled and dualizable categories

3. Examples

4. Compact morphisms and AB6

5. Calkin categories and colimits of dualizable categories

6. Limits of dualizable categories

7. Semiorthogonal decompositions and infinite products

8. Flat presentable categories

9. Localizing invariants of dualizable categories

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